

Erdős Problem #959

EP-959

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $A \subset \mathbb{R}^2$ be a set of size n and let $\{d_1, \dots, d_k\}$ be the set of distinct distances determined by A . Let $f(d)$ be the number of times the distance d is determined, and suppose the d_i are ordered such that $f(d_1) \geq f(d_2) \geq \dots \geq f(d_k)$. Estimate $\max(f(d_1) - f(d_2))$, where the maximum is taken over all A of size n .

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>